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Visualisation-VF.CONCEPT

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1 Introduction

This document is a function specification for the control of the Visualisation system for V300ZEFIRO for customer Trenitalia (VZES Project).

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

ACM	Auxiliary Converter Module
CD	Circuit Diagram
CCU	Central Computing Unit
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FCM	MF-DC Converter Module
FPGA	Field-Programmable Gate Array
HMI	Human Machine Interface
HW	Hardware
ID	Identifier
IDU	Interface Display Unit
IPTCom	Internet Protocol Train Communication
LCB	Line Circuit Breaker
LCM	Line Converter Module
LED	Light Emitting Diode
LIM	Line circuit protection and Interface Monitor
MCB	Mini Circuit Breaker
MCM	Motor Converter Module
PD	Process Data
SDT	Safe Data Telegram
SIF	Safety Instrumented Function
SIL	Safety Integrity Level
TCMS	Train Control and Management System
TDS	Train Diagnostics System
WSP	Wheel Slide Protection

2 System Overview

The Visualisation system consists of the TCMS IDUs in the driver's cabs and the crew compartment.

The purpose of the TCMS IDU, hereafter called IDU, is to provide a graphical user interface to the TCMS. This document describes the functional requirements on the IDU. The IDU Picture model describes each menu graphically, see [1].

2.1 System Configuration

There are two IDUs located in each cab and one IDU located in the crew compartment in the M3 car. In the cab the primary IDU is located in front of the driver and the back-up IDU is located to the left of the driver. The back-up IDU is mounted behind a transparent cover and is not possible to touch. If the primary IDU fails the driver can replace the primary IDU unit with the back-up IDU unit when the train is at standstill.

2.2 Normal condition

In normal condition mode the primary IDU in the cab is the activated one, i.e. the IDU for the Driver, Maintainer or Crew to interact with. The secondary back-up IDU is in idle mode at this condition.

2.3 Failure condition

In failure condition mode the secondary IDU in the cab, i.e. the backup IDU, is activated. It is then the user's responsibility to move it from the secondary location to the primary location. Once the switch has been performed and the IDU is powered up it is ready to be used.

2.4 Failure supervision

Failure supervision is performed either by the user, seeing that the IDU has become blank or in another abnormal state, or by the lifesign facility between the IDU and train computer.

2.5 Safety concept

The TCMS Platform and its safety products, e.g. HMI 411-S, are capable of implementing a set of generic safety functions. Our application will utilize the following generic safety functions:

- Display safety related data (SIF1).

- HMI confirmation with hard key (SIF7).

There are two main components in the HMI411-S unit, main CPU and safety CPU.
The main CPU executes the IDU application:

- Receive SDTv2 process data over external Ethernet connector.
- Display Safe Components on screen. These are graphical representations of safe variables extracted from SDTv2 process data.

The safety CPU perform supervision of the IDU application based on configuration.

- Extract and validate SDTv2 process data received from the main CPU.
- Supervise that the main CPU correctly displays Safe Components according to SDTv2 process data.
- Process button presses on Safe hardkeys, used to confirm SDTv2 process data

The TCMS Platform and its safety products implement two distinct types of safe states. A non operational safe state which is specific for each product and an operational safe state which is specific for each safety function.

- Non operational Safe State
  When the HMI-411 enters non operational safe state, the power supply to almost all hardware parts are turned off. This means that the unit won't display any graphics, it is completely silent and no network communication will take place. The only thing that may happen in this state, is that the status LED will blink with blue light for diagnostic purposes. The only way to recover from safe state is to toggle the external power supply to the unit.

- Operational Safe State
  The HMI-411 implements 8 different operational safe states. One associated with each one of the safety functions. A brief description of the safety functions utilized in the Zefiro IDU, its operational safe state and the recovery condition will be presented in the following sections.

2.5.1 Display of safe data

- Operational safe state
  For display of safe data, SDTv2 process data updates are required to come in between certain time intervals. If the communication is lost or if the data is considered invalid; this safety function shall be enforced into blank state. In that case, the main processor application shall display the blank state graphics instead of the safe data representation. This can be any user configured static graphical object, like a solid color or a cross.
  The outgoing status telegram will reflect that operational safe state was enforced for display of safe data.
  The status LED will also signal to the train driver that this function is unavailable.
  If the main processor application fails to display the blank state graphics when required, non operational safe state will be enforced.
- Recovery condition
  When in operational safe state, at least two consecutive safe data updates must be received. If they are considered valid, the safe data representation will resume the visualization.

2.5.2 Safe confirm with hard key

- Operational safe state
  For safe confirm with hard key, a page configured for safe confirmation shall be displayed. That specific page shall present safe data using display of safe data. On the same page one hardkey on the front panel shall be configured as the active confirm button. If there is an error on that specific button, any presses/releases will be ignored.
  The outgoing status telegram will reflect that operational safe state was enforced for safe confirm with hard key.
  The status LED will also signal to the user that this function is unavailable.
- Recovery condition
  The button will be re-checked again every minute. If it has recovered it will be possible to confirm the data by pressing the active confirm button.

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3 Functional requirements

3.1 General

3.1.1 Graphical Strategy for IDU

3.1.1.1 FunctionDesign

To provide a consistent overall look & feel of the HMI a standard screen layout is used. The standard screen always covers the whole screen, 800 x 600 pixels.
The standard screen layout includes the following fields:

- Header field
- Title field
- Active content field
- Event field
- Footer field

Most menus will incorporate all fields in the standard screen layout. Under specific circumstances, i.e. lack of screen space or desire to show as little information as possible circumstances, certain fields may not be used. This presents an opportunity to enlarge other aspects of the screen, namely the active content field.

The contents of the header field, title field and footer field are specified in the requirements below. The event field is specified in chapter Events. The contents of the active content field is specified for each menu together with any exceptions from the standard screen layout.

3.1.1.2 Requirements

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3.1.2 Audible message

3.1.2.1 Function Design

When an event has occurred that requires the train staff attention there will be an audible message or tone to direct the train staff to the IDU in the crew compartment. An in all car audible message shall be used to get the crew's attention.

3.1.2.2 Requirements

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3.1.4.2 Requirements

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The same menu structure is used as a base for all user groups and locations. The principle is that when a function is not available the corresponding menu is removed keeping the functions geographically consistent in all menu trees.

The menu structure has been designed to be shallow, allowing quick access to functionality that requires it, while not placing too many functions on a single page, avoiding clutter. Shortcuts are available for important and frequently used pages to speed up the navigation process.

Two different types of navigation are present in the IDU application. The two types are manual navigation and forced navigation.

- Manual navigation is navigation performed by the user, i.e. pressing a navigation button on the IDU display which performs a menu change.
- Forced navigation is navigation caused by a system state, e.g. when decoupling has been performed the IDU application automatically displays the Decoupling system message menu.

The forced navigation has a priority for deciding of which menu to show.

3.2.2 Requirements

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3.3 Menus

3.3.1 Primary IDU menus

3.3.1.1 Login menu

3.3.1.1.1 Function Design

To access functionality of the IDU a login ID is required. With login ID different user groups and languages are distinguished. The user groups are:

- Driver
- Crew
- Maintainer

The supported languages are:

- Italian
- French
- Spanish
- English

When a user logs in, the language that is defined for that user is displayed.

The language displayed can be changed through the Settings menu.

3.3.1.1.2 Requirements

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Serious fault events have the highest priority and takes precedence over all other category of events even if an event of another category has already been announced.

Advisory events have the second highest priority, and takes precedence over Alarm and Warning events even if an Alarm or Warning already has been announced.

Alarm events have the third highest priority and takes precedence over Warning events even if a Warning already has been announced.

3.3.2.2 Active event overview menu

3.3.2.2.1 Function Design

The active event overview menu provides a clear overview of the systems on the train that have active events. It also provides navigation possibilities to select systems with active events. With navigation it is meant that the active event list is shown, and contains active events for selected system.

3.3.2.2.2 Requirements

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3.3.2.4 Event history overview menu

3.3.2.4.1 Function Design

The event history overview menu is used to make a selection to only see history events for a specific event group.

3.3.2.4.2 Requirements

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3.3.2.7.3 Serious fault events

3.3.2.7.3.1 Function Design

When a Serious fault event has occurred the event is announced on the IDU to give added information to the serious fault that has occurred.

The Serious fault event is only announced on the IDU in active cab and it needs to be acknowledged. The event is available in the event history menu, and in the active event list if the serious fault event is still active.

The Serious fault event is presented in the event field of the IDU. If the event field is pressed, the Expanded event menu is shown presenting the user more information about the Serious fault event.

Note. Serious fault is acknowledged using the button on the frame of the IDU, see chapter 3.3.2.8. The serious fault event that is stored in the ODBS in parallel is also acknowledge, after the user has read it, using the acknowledge button on the IDU display.

3.3.2.7.3.2 Requirements

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3.3.2.7.6 Warning events

3.3.2.7.6.1 Function Design

When a Warning has occurred immediate action has to be taken by the user, due to a train malfunction or impact on train operation that is limited. The Warning event is announced on the IDU in active cab and/or in the crew compartment. The Warning events need to be acknowledged by the user.

The event is available in the event history menu, and in the active event list if the warning is still active.

The Warning event is presented in the event field of the IDU. If the Warning in the event field is pressed, the Expanded event menu is shown presenting the user more information about the Warning event.

3.3.2.7.6.2 Requirements

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3.3.2.9.6.2 Requirements

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3.3.2.9.9 Waiting for train configuration

3.3.2.9.9.1 Function design

When reconfiguration of the train topology has occurred and configuration is not valid the Waiting for train configuration system message is shown on IDU.

3.3.2.9.9.2 Requirements

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3.3.3.4 High voltage system menu

3.3.3.4.1 Function Design

The High voltage system menu provides functionality related to train's high voltage systems. E.g. circuit breakers and disconnector status.

3.3.3.4.2 Requirements

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3.3.3.6.2 Requirements

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3.3.3.9.2 Requirements

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3.3.3.12 Air supply system menu

3.3.3.12.1 Function Design

The Air supply system menu provides functionality related to train's air supply systems. E.g. status of compressors and sensors, and performing cut out/cut in of compressors.

3.3.3.12.2 Requirements

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3.3.4 Menus for Settings

3.3.4.1 Settings menu

3.3.4.1.1 Function Design

The Settings menu provides the user with settings related functionality. From this menu the user can proceed to e.g. the language selection menu and perform logout action.

3.3.4.1.2 Requirements

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

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[1] PGR_3EST000225-7920 V300Zefiro IDU Picture model

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6 Revision Log

6.1 Revision log 3ECP413922-1003__

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Date:
Editor:

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